Physics - 2105

Assignment - 01

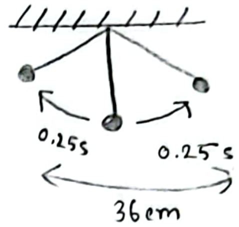
Md. Mustiqur Rahman

ID: 011 221 334

Section : C

Submitted to : Sagar Dutta

① An object travel from one point of zero velocity to the reset such ·point, it has complated one half of a full oscillation.



① Period $= 0.25 + 0.25 = 0.5$ s

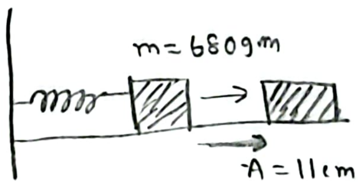
(ii) Frequency $f = \frac{1}{T} = \frac{1}{0.5} = 2$ Hz

(iii) mean to maximum displacement is called to Amplitude

$x = 0.36$ cm

$A = \frac{x}{2} = \frac{0.36}{2} = 0.18$ m.

② given,



$m = 680$ gm $= 0.68$ kg

$K = 65$ N/m

$A = 11$ cm $= 0.11$ m

(ii) Angular frequency $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{65}{0.68}} = 9.78$ rad/s

① Time period $T = \frac{2\pi}{\omega} = \frac{2\pi}{9.78} = 0.64$ sec.

(iii) Phase constant, $\phi = 0°$

(IV) velocity $v_1 = -A\omega \sin(\omega t)$ $A = 0.11m$

$= -.11 \times 9.78 \sin(9.78 \times 4)$ $t = 4s$

$= -\cancel{1.068}\ \cancel{ms^{-1}}$ $= -1.063 ms^{-1}$

$= \cancel{-0.678 ms^{-1}}$

$V_{max} = -A\omega$

$= -1.07\ ms^{-1}$

(V) acceleration, $t = 4$

$a = -A\omega^2 \cos(\omega t)$

$= -0.11 \times (9.78)^2 \cos(9.78 \times 4)$

$= -1.57\ ms^{-2}$

$a_{max} = -A\omega^2$

$= -10.52\ ms^{-2}$

(VI) displacement.

$\cancel{x = 0.11 m}$ $t = 0$

$A \cos(\omega t)$

$= 0.11 \times \cos(9.78 \times 0)$

$= 0.11\ m$

$t = 7 sec$

$x = 0.11 \times \cos(9.78 \times 7)$

$= 0.08 m.$

(VII) velocity at displacement.

$x = 0.11 m$

$v = \omega\sqrt{A^2 - x^2}$

$= 0\ ms^{-1}$

$x = 0.04 m$

$v = \omega\sqrt{(0.11)^2 - (0.04)^2}$

$= 1.002\ ms^{-1}$

(3) Given,

$m = 0.12\,kg$

$A = 8.5\,cm = 0.085\,m$

$T = 0.20\,s$

(i) $a = -A\omega^2$

$= -A\left(\frac{2\pi}{T}\right)^2$

$= -83.89\ ms^{-1}$

$F = ma.$

$= 0.12 \times 83.89$

$= 10\ N$

(ii) $\omega^2 = \frac{k}{m}$

$K = \omega^2 m$

$= \left(\left(\frac{2\pi}{T}\right)^2 \times 0.12\right)$

$= 118.44\ N/m$

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(4) given = $x = 2.00\ mm$

$= 2 \times 10^{-3}\ m$

$f = 120\,Hz$

(i) Amplitude $A = \frac{x}{2} = \frac{2\times10^{-3}}{2}$

$= 1\times10^{-3}\ m$

(ii) maximum blade speed $V_{max} = \omega A$

$= 2\pi f.\,A$

$= (2\pi \times 120)\times 10^{-3}$

$= 0.75\ ms^{-1}$

(vii) maximum acceleration of blad $a_{max} = \omega^2 A$

$= (2\pi \times 120) \times 1\times10^{-3}$

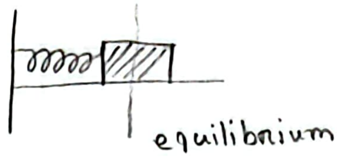
$= 568.49\ ms^{-2}$

⑤ given $k = 400\ N/m$

$x = 0.100\ m$

$v = -13.6\ ms^{-1}$

$a = -123\ ms^{-2}$



(i) we know, $a = -\omega^2 x$

$$\therefore \omega = \sqrt{\frac{a}{-x}}$$

$$= \sqrt{\frac{-123}{0.100}}$$

$$= 35.07\ rad/s$$

$$\therefore f = \frac{\omega}{2\pi} = \frac{35.07}{2\pi} = 5.58\ Hz$$

(ii) we know,

$$\omega^2 = \frac{k}{m}$$

$$\therefore m = \frac{k}{\omega^2}$$

$$= \frac{400}{(35.07)^2} = 0.325\ kg$$

(iii) $\boxed{v = \omega\sqrt{A^2 - x^2}}$

$$v^2 = \omega^2 (A^2 - x^2)$$

$$\omega^2 (A^2 - x^2) = v^2$$

$$A^2 - x^2 = \frac{v^2}{\omega^2}.$$

$$A^2 = \frac{v^2}{\omega^2} + x^2$$

$$A = \sqrt{\frac{v^2}{\omega^2} + x^2}$$

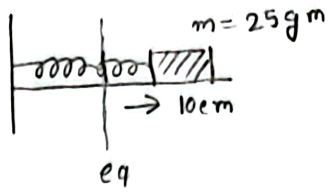
$$= \sqrt{\frac{(-13.6)}{(35.07)^2} + (0.1)^2} = 0.400\ m.$$

(6) Given,

$m = 25\ gm = 0.025\ kg$

$k = 400\ dyns/cm = 0.4\ N/m$

$x = 10\ cm = 0.1\ m = A$



(iii) angular frequency $\omega = \sqrt{\frac{k}{m}}$

$$= \sqrt{\frac{0.4}{0.025}}$$

$$= 4\ rad/s$$

(ii) Frequency, $f = \frac{\omega}{2\pi} = \frac{4}{2\pi} = 0.636\ Hz$

(i) Time period, $T = \frac{1}{f} = \frac{1}{0.636} = 1.57s.$

(iv) maximum velocity $V_{max} = A\omega = 0.1 \times 4$

$$= 0.40\ ms^{-1}$$

(7) given, $m = 1.68 \times 10^{-27}\ kg$

$f = 10^{14}\ Hz$

$A = 10^{-10}\ m$

$\omega = 2\pi f = 2\pi \times 10^{14}\ rad/s$

$a_{max} = \omega^2 A$

$$= (2\pi \times 10^{14})^2 \times 10^{-10}$$

$$= 3.947 \times 10^{19}$$

$\therefore F_{max} = m a_{max}$

$$= 1.68 \times 10^{-27} \times 3.947 \times 10^{19} = 6.63 \times 10^{-8}\ N.\ (Am)$$

⑧ given,

$V_{max} = 1\ ms^{-1}$

$a_{max} = 1.57\ ms^{-2}$

we know,

$V_{max} = \omega A$ — ①

$a_{max} = \omega^2 A$ — ⑪

⑪ ÷ ①

$$\frac{a_{max}}{V_{max}} = \frac{\omega^2 A}{\omega A}$$

$$\frac{1.57}{1} = \omega$$

$\therefore \omega = 1.57\ rad/s$

$$\therefore T = \frac{2\pi}{\omega} = \frac{2\pi}{1.57} = 4\ sec.$$

⑨ given,

$A = 5m$

$x = 3m$

$a = 48\ ms^{-2}$

① we know,

$a = -\omega^2 x$

$$\omega = \sqrt{\frac{-48}{3}} = 4\ rad/s$$

$$\therefore v = \omega\sqrt{A^2 - x^2} = 4\sqrt{5^2 - 3^2} = 16\ ms^{-1}$$

⑪ $T = \frac{2\pi}{\omega} = \frac{2\pi}{4} = 1.57s.$

⑪ maximum valocity $V_{max} = A\omega$

$= 5 \times 4$

$= 20\ ms^{-1}$

(10)

Given,

$x = 0.37 cm$

$v = 0 m/s$

$f = 0.25 Hz.$

(i) Time period, $T = \frac{1}{f} = \frac{1}{0.25} = 4s$

(ii) angular frequency $\omega = 2\pi f = 2\pi \times 0.25 = 1.57 rad/s$

(iii) $A = x = 0.037 cm$ [$\because v = 0;$]

$= 0.0037 m$

(iv) displacement. $\boxed{x = A \cos \omega t}$

$\Rightarrow x = 0.0037 \cos(1.57t)$

$\Rightarrow x_{(3s)} = 0.0037 \times \cos(1.57 \times 3)$

$= -8.83 \times 10^{-6} m$

(v) Velocity $v = \frac{dx}{dt} = -(0.0037 \times 1.57) \sin(1.57 \times 3)$

$= 5.97 \times 10^{-3} m/s$

(11) given,

$$x = 10\cos\left(3\pi t + \frac{\pi}{3}\right)$$

(i) $t = 2.5$

$$x = 10\cos\left(3\pi \times 2.5 + \frac{\pi}{3}\right)$$

$$= 8.66\ m.$$

(ii) · velocity $= \dfrac{dx}{dt}$

$$= \frac{d}{dt}\left(10\cos\left(3\pi t + \frac{\pi}{3}\right)\right)$$

$$= -10\sin\left(3\pi t + \frac{\pi}{3}\right) \cdot \frac{d}{dt}\left(3\pi t + \frac{\pi}{3}\right)$$

$$= -10\sin\left(3\pi t + \frac{\pi}{3}\right) \times 3\pi$$

$$= -10(3\pi)\sin\left(3\pi t + \frac{\pi}{3}\right)$$

$$= -10(3\pi)\sin\left(3\pi \times 3 + \frac{\pi}{3}\right)$$

$$= 81.62\ ms^{-1}$$

(iii) acceleration, $a = -A\omega^2 \cos(\omega t + \phi)$

$$= -10(3\pi)^2 \cos\left(3\pi \times 2 + \frac{\pi}{3}\right)$$

$$= -444.13\ ms^{-2}$$

(iv) maximum velocity, $V_{max} = -\omega A$

$$= -3\pi \times 10$$

$$= -94.24\ ms^{-1}$$

(12) Given, $x = 10 \sin(10t - \frac{\pi}{6})$

$A = 10\,m$

$\omega = 10$ rad/s

$\phi = -\pi/6$

(i) $f = \frac{\omega}{2\pi} = \frac{10}{2\pi}$

$= 1.59\ Hz$

(ii) Time period, $T = \frac{1}{f} = \frac{1}{1.59}$

$= 0.628\,s.$

(iii) maximum displacement, $A = 10\,m$

(iv) $V_{max} = A\omega$

$= 10 \times 10$

$= 100\ ms^{-1}$

(v) maximum acceleration, $a_{max} = -A\omega^2$

$= -10^2 \times 10$

$= -1000\ ms^{-2}$

(13) Given, $A = 4.00\,m$

$f = 0.5\ Hz$

$\phi = \pi/4$

(i) $T = \frac{1}{f} = \frac{1}{.5} = 2$ sec.

(ii) $\omega = 2\pi f = 2\pi \times 0.5 = \pi.$

We know, $x = A\cos(\omega t + \phi)$

$\therefore x = 4\cos(\pi t + \frac{\pi}{4})$

(iii) From, (ii)

$$x = 4\cos\left(\pi t + \frac{\pi}{4}\right).$$

$\therefore$ velocity $v = \dfrac{dx}{dt}$

$$= \frac{d}{dt}\left(4\cos\left(\pi t + \frac{\pi}{4}\right)\right)$$

$$= -4\sin\left(\pi t + \frac{\pi}{4}\right) \times \pi$$

$$= -4\pi \sin\left(\pi t + \frac{\pi}{4}\right)$$

now, $V_{(5)} = -4\pi \sin\left(\pi \times 5 + \frac{\pi}{4}\right)$

$$= 8.89 \text{ ms}^{-1}$$

accelaration, $a = \dfrac{d\cancel{x}}{dt}$

$$= -A\omega^2 \cos(\omega t + \phi)$$

$$= -4(\pi)^2 \cos\left(\pi t + \frac{\pi}{4}\right)$$

$$= 27.915 \text{ ms}^{-2}$$

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(14) Given,

$m = 2\text{kg}$

$k = 196 \text{ N/m}$

(i) $A = 5\text{cm} = 0.05\text{m}$

(i) $W = \sqrt{\dfrac{k}{m}} = \sqrt{\dfrac{196}{2}} = 9.9$ rad/s

(ii) $\omega = 2\pi f$

$\therefore f = \dfrac{\omega}{2\pi} = \dfrac{9.9}{2\pi} = 1.57$ Hz

(iii) $T = \dfrac{1}{f} = \dfrac{1}{1.57} = 0.636$ s.

(ii) From the question I can write the equation of displacement

is $x = A\cos(\omega t + \phi)$

$\therefore$ for x Vs t

displacement $x_{(t)} = 5\cos(1.57t)$ $[\phi = 0$ because $t = 0]$

(15)

(i)

$F = |-kx|$

$mg = |-kx|$

$k = \frac{mg}{x}$

$= \frac{0.5 \times 9.8}{0.07} = 70\ N/m$

(ii) $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{70}{0.5}} = 11.832\ rad/s$

(iii) $T = 2\pi/\omega = \frac{2\pi}{11.832} = 0.531$ sec.

(iv) Initial Potential energy -

$E_P = \frac{1}{2}kx^2$

$= \frac{1}{2} \times 70 \times (0.07)^2$

$= 0.1715\ J$.

(v) kinitie energy, $E_K = \frac{1}{2}mv^2$

$= \frac{1}{2}(0.5) \times (0.4)^2$

$= 0.04\ J$

(vi)

(16) Given, $m = 25gm = 0.025\,kg$ $\quad v = 40\,cm/s$

$= 0.4\,ms^{-1}$

$k = 400\,dynes/s = 0.4\,N/m$

$x = 10\,cm = 0.1\,m$

(i) $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{0.4}{0.025}} = 4\,rad/s$

$\therefore T = \frac{2\pi}{\omega} = \frac{2\pi}{4} = 1.57\,sec.$

(ii) Frequency $f = \frac{1}{T} = \frac{1}{1.57} = 0.636\,Hz$

(iii) angular frequency from (i) $\omega = 4\,rad/s.$

(iv) Total energy $E_T = E_P + E_K$

$\therefore E_P = \frac{1}{2}kx^2$

$= \frac{1}{2} \times 0.4 \times (0.1)^2$

$= 0.002\,J$

$E_K = \frac{1}{2}mv^2$

$= \frac{1}{2}(0.025) \times (0.4)^2$

$= 0.002\,J$

$\therefore E_T = 2\times10^{-3} + 2\times10^{-3}$

$= 4\times10^{-3}\,J.$

(e) from (iv) $E_T = 4\times10^{-3}$ J

we, know,

$$E_T = \frac{1}{2}kA^2$$

$$A^2 = 2E_T/k$$

$$\Rightarrow A = \sqrt{\frac{2E_T}{k}}$$

$$= \sqrt{\frac{2\times4\times10^{-3}}{0.4}}$$

$$= 0.141 m$$

(vi) $x = A\cos\theta$ $[\theta_0 = \omega t + \phi]$

$$\Rightarrow 0.1 = 0.14\cos\theta_0$$

$$\therefore \theta_0 = 44.4°$$

(vii) $V_{max} = -A\omega$

$$= -0.141\times4$$

$$= 0.565\ ms^{-1}$$

(viii) $a_{max} = -A\omega^2$

$$= -0.141(4)^2$$

$$= -2.256\ ms^{-2}$$

(ix) Given,

from (v) → $A = 0.141\,m$

$\omega = 4\,rad/s$

$x = A\cos(\omega t)$

$= 0.141\cos(4t)$

$V_{(t)} = -A\omega\sin(\omega t)$

$= -0.141 \times 4 \times \sin(4t)$

$a_{(t)} = -A\omega^2\cos(\omega t)$

$= -0.141 \times (4)^2\cos(4t)$

$= -2.256\cos(4t)$

(x) Given, $t = \frac{\pi}{8}$

$x = 0.141\cos(4 \times \frac{\pi}{8})$

$= 0\,m$

$V_{(t)} = -0.141 \times 4 \times \sin(4 \times \frac{\pi}{8})$

$= -0.564\ ms^{-1}$

$a_{(t)} = -2.256\cos(4 \times \frac{\pi}{8})$

$= 0\ ms^{-2}$

(17) Given,

$m = 2.72 \times 10^5$ kg

$f = 10 Hz$.

(i)

$A / x_{max} = 20 cm = 0.2 m$

$\omega^2 = \frac{K}{m}$

$K = \omega^2 m$

$= (2\pi f)^2 \times m$

$= (2\pi)^2 \times 2.72 \times 10^5$

$= 1.0738 \times 10^9$

mechanical energy $= E_T = \frac{1}{2} k A^2$

$= \frac{1}{2} \times 1.0738 \times 10^9 \times (0.2)^2$

$= 21.48 \times 10^6$ J

(ii) At equilibrium point. $E_P = 0$

$E_K = \frac{1}{2} kA^2 = E_T$

or, $E_K = \frac{1}{2} mv^2$

$\therefore E_T = 0 + \frac{1}{2} mv^2$

$v = \sqrt{\frac{2E_T}{m}}$

$= \sqrt{\frac{2 \times 21.48 \times 10^6}{2.72 \times 10^5}}$

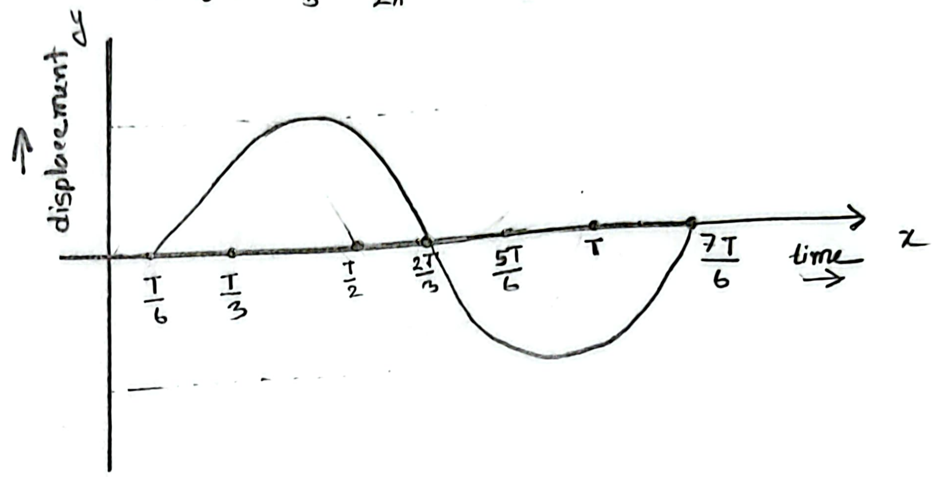
$= 12.567 \; ms^{-1}$

(18) (i) $y = A\sin\left(\omega t - \frac{\pi}{3}\right)$

$\omega t - \frac{\pi}{3} = 0$

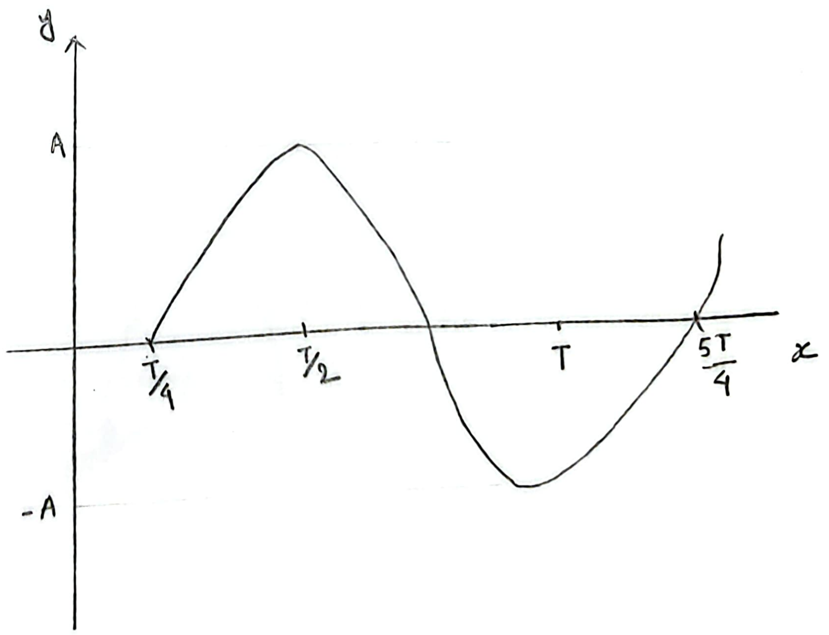
$\omega t = \frac{\pi}{3} \rightarrow \frac{2\pi}{T} t = \frac{\pi}{3}$

$\Rightarrow t = \frac{\pi}{3} \times \frac{T}{2\pi} = \frac{T}{6}$



(ii) $y = A\sin\left(\omega t - \frac{\pi}{2}\right)$

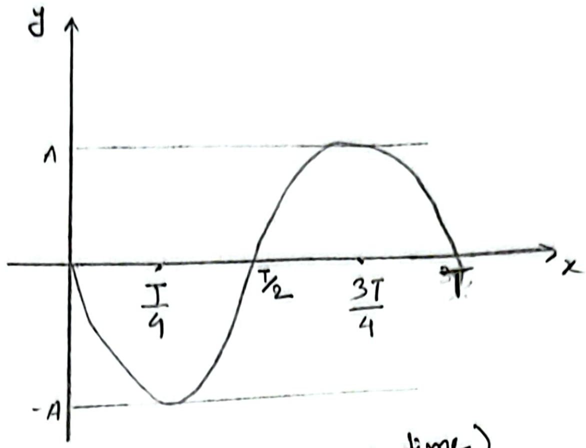
$\omega t = \frac{\pi}{2} \quad \therefore t = \frac{\pi}{2} \times \frac{T}{2\pi} = \frac{T}{4}$



displacement(y) vs time(x)

(iii) $y = A \sin (\omega t + \pi)$

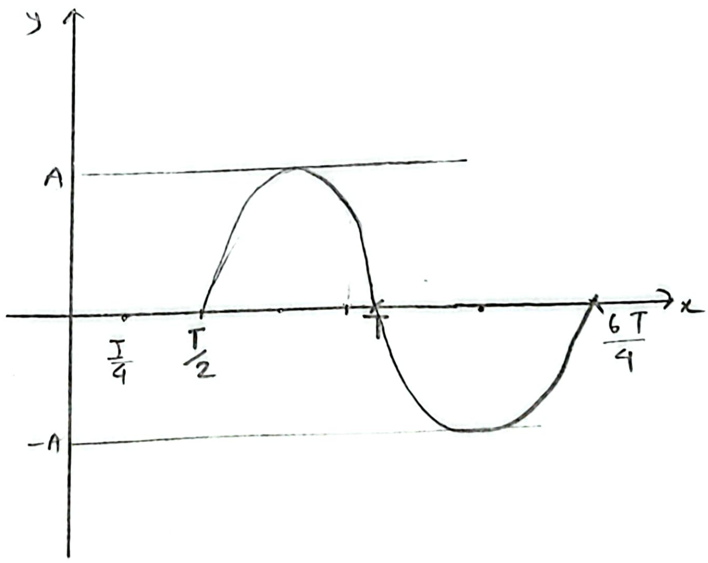
$\omega t = -\pi$, $\therefore t = -\pi \times \frac{T}{2\pi} = -\frac{T}{2}$



(displacement vs time)

(iv) $y = A \sin (\omega t - \pi)$

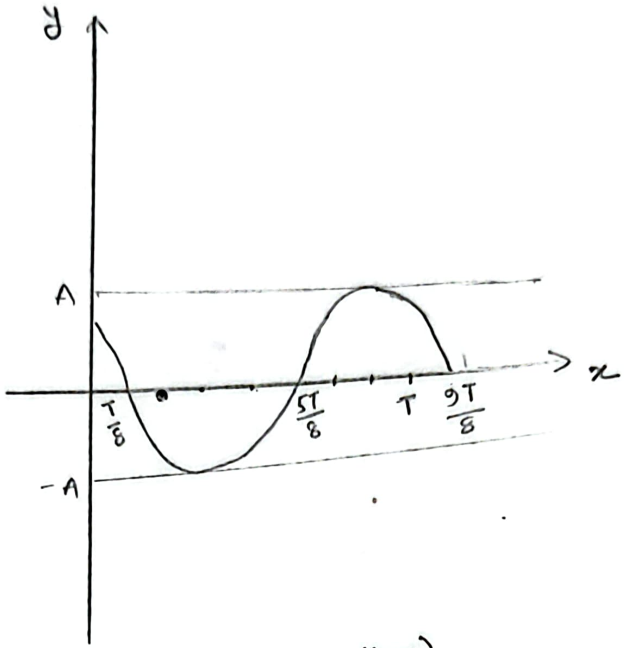
$\omega = \pi$ $\therefore t = \pi \times \frac{T}{2\pi} = \frac{T}{2}$



(displacement vs time.)

$y = A\sin\left(\omega t + \frac{3\pi}{4}\right)$

$\omega = -\frac{3\pi}{4}$, $\therefore t = -\frac{3\pi}{4} \times \frac{T}{2\pi} = -\frac{3T}{8}$



(displace ment vs time)